



Model Capacity and Inductive Bias

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Problem Formulation

Outline

Linear Regression

Logistic
Regression and
Softmax
Regression

Overfitting,
Underfitting, and
Bias-Variance

Feature
Transformation:
The Missing Piece

Representation
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Outline

- ➊ Linear Regression
- ➋ Logistic Regression and Softmax Regression
- ➌ Overfitting, Underfitting, and Bias–Variance
- ➍ Feature Transformation: The Missing Piece
- ➎ Representation Learning



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Linear Regression

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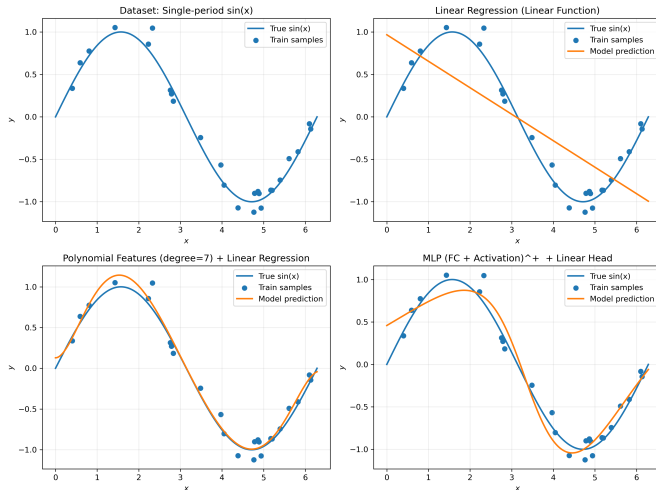
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Model Capacity

What Can Linear Regression Learn?



Same dataset, different representations $\Rightarrow$ different function approximations.

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What Can Linear Regression Learn?



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Key Message

- Linear regression is a **linear function approximator**.
- Prediction is always an affine function of the input:

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b$$

Linear regression cannot model non-linear target functions in the original input space.

Model Capacity

Geometric Interpretation



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Function Space View

- Linear regression fits a function in a **linear hypothesis space**.
- For 1D input: predicted curve is always a **straight line**.
- For higher dimensions: prediction is a **hyperplane**.

Model capacity is limited by the dimensionality and linearity of the feature space.

Model Capacity

Implications

Limitations

- Cannot represent:
 - Curved functions (e.g. $\sin x$, x^2)
 - Periodic patterns
 - Piecewise or highly non-linear relationships

The limitation comes from the linear form, not from the loss function.

Key Idea

- Linear regression is **always linear in the feature space**.
- Next: we need feature transformation.



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Logistic Regression and Softmax Regression

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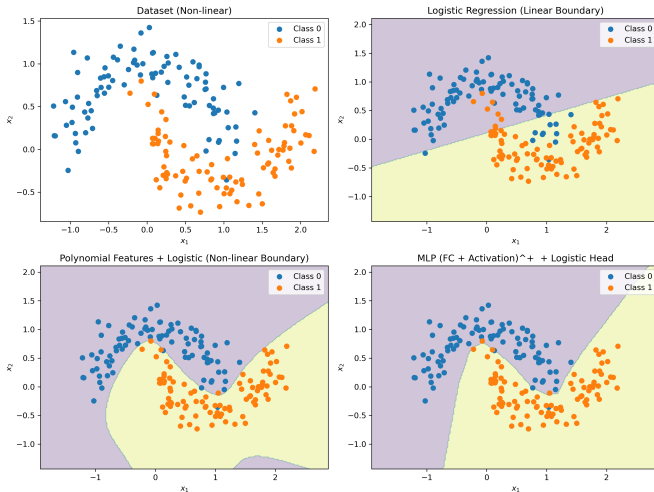
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What Can Logistic Regression Learn?



Same dataset, different representations $\Rightarrow$ different decision boundaries.

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What Can Logistic Regression Learn?



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Key Message

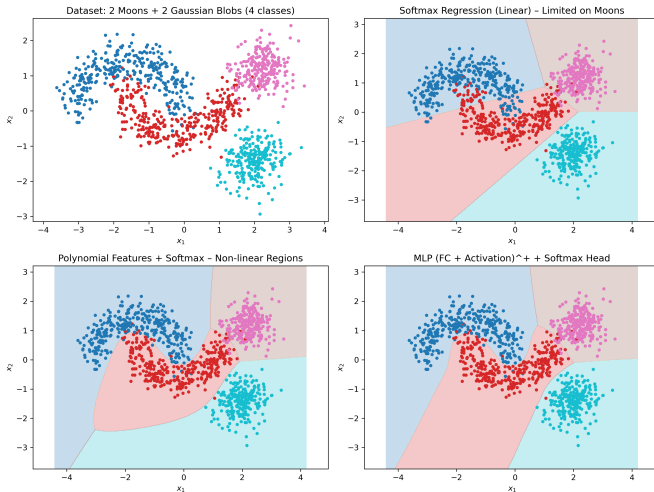
- Logistic regression is a **linear classifier** in the input feature space.
- Decision boundary is always a hyperplane:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

Logistic regression itself cannot model non-linear decision boundaries.

Model Capacity

What Can Softmax Regression Learn?



Same dataset, different representations $\Rightarrow$ different decision boundaries.

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What Can Softmax Regression Learn?



Key Message

- Softmax regression is a **linear classifier for multi-class problems**.
- Each class has its own linear score function:

$$z_k = \mathbf{w}_k^T \mathbf{x} + b_k$$

- Decision boundaries between any two classes are linear:

$$(\mathbf{w}_i - \mathbf{w}_j)^T \mathbf{x} + (b_i - b_j) = 0$$

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What Can Softmax Regression Learn?



Implications

- Class regions are **convex polytopes**
- Boundaries are combinations of hyperplanes
- Cannot represent curved or disconnected class regions

Softmax regression generalizes logistic regression to multi-class settings, but remains linear in feature space.

Key Idea

- Logistic / Softmax regression are **always linear**
- Next: we need feature transformation

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Overfitting, Underfitting, and Bias–Variance

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**Overfitting,
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Overfitting vs Underfitting

Intuition

Underfitting

Model is too simple; cannot capture the true pattern in the data.

- High training error and high validation error
- Example: linear model for a curved relationship

Overfitting

Model is too complex; memorizes training data including noise.

- Low training error but high validation error
- Example: high-degree polynomial that wiggles through every training point

Goal

Find a model complexity that balances both: good generalization to unseen data.



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Bias–Variance Tradeoff

Decomposition of Error



Decomposition

Expected prediction error can be decomposed as:

$$\mathbb{E}[(\hat{y} - y)^2] = \underbrace{(\mathbb{E}[\hat{y}] - y)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\hat{y} - \mathbb{E}[\hat{y}])^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible noise}}$$

- **Bias:** error from wrong assumptions; underfitting $\Rightarrow$ high bias
- **Variance:** sensitivity to the training sample; overfitting $\Rightarrow$ high variance
- **Trade-off:** increasing model capacity reduces bias but increases variance

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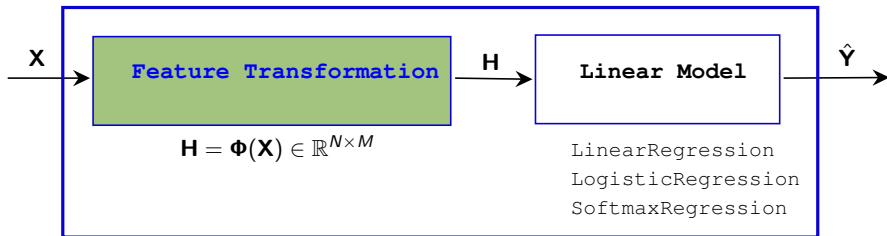
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Feature Transformation

From Linear Models to Nonlinearity



- Linear models are powerful only in the feature space they operate on.
- Nonlinearity comes from the representation: $\Phi(\mathbf{X})$.
- This is the core idea behind modern deep learning.



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Why Linear Models Fail on Nonlinear Patterns

A Linear Model is Always Linear in Its Inputs

$\hat{y} = \mathbf{w}^T \mathbf{x} + b \Rightarrow$ hyperplane decision boundary / straight-line regression

The Bottleneck

- The limitation is not the optimizer, not the dataset size.
- The limitation is the hypothesis class: **linear functions in the current feature space.**

To learn nonlinear relations, we must change the feature space.



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The Core Trick: Linear in $\Phi(\mathbf{x})$

Feature Mapping

Define a mapping:

$$\Phi : \mathbb{R}^D \rightarrow \mathbb{R}^M$$

Then apply a linear model:

$$\hat{y} = \mathbf{w}^T \Phi(\mathbf{x}) + b$$

Interpretation

- The model is still linear in parameters $(\mathbf{w}, b)$.
- But it can be **nonlinear in the original input $\mathbf{x}$** .

This is the foundation of basis expansion, kernels, and deep networks.

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Feature Transformation

Three Families of Feature Transformations

1) Manual Feature Transform

- Polynomial features, Fourier basis
- RBF features, kernels (implicit Φ)

2) Shallow Learned Features

- MLP: $(\text{FC} + \text{Nonlinearity})^+$ as Φ

3) Deep Learned Features

- CNN (2012), RNN, GNN
- Transformer (2017), Mamba (2023–)



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Bridge to the Next Chapters



Big Picture

- Modern models differ mainly in how they design and learn Φ .
- The head is often simple (linear classifier / regressor).

What Comes Next

- MLP as the simplest learned feature transformation
- Loss functions and optimization
- CNN / Transformer / Mamba as advanced Φ modules

Next: we start with MLP as a learnable feature transformer.

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Representation Learning

Core Idea

What is Representation Learning?

- Learning a transformation

$$\Phi(x)$$

that maps raw input to a **useful representation**

- The representation makes downstream tasks easier

Unified View

$$x \xrightarrow{\Phi} \text{representation} \xrightarrow{g(\cdot)} \hat{y}$$

- Φ : feature / representation transformation
- g : simple predictor (often linear)

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Historical Perspective

Before Deep Learning

- Humans design features manually
- Examples:
 - Polynomial features
 - SIFT, HOG
 - Domain-specific descriptors

With Deep Learning

- Features are **learned automatically** from data
- Neural networks learn Φ directly

Deep learning replaces manual feature engineering with learned representations.



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Representation Learning

Connection to Deep Learning

Key Observation

- Deep networks = many stacked transformations

$$\Phi(\mathbf{x}) = f_L(\cdots f_2(f_1(\mathbf{x})))$$

- Each layer refines the representation

Interpretation

- CNN: learn spatial representations
- RNN: learn temporal representations
- Transformer: learn contextual representations
- Mamba / SSM: learn sequence state representations

Deep learning = deep representation learning.

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